Testing and Deployment

EduGenie - AI Learning Assistant

1. Introduction

This document explains how EduGenie is tested and how it is deployed. EduGenie depends on a local model, a cloud API, a web server and a browser page, so testing checks each layer separately and then the whole flow. Deployment makes sure the API key, the heavy PyTorch dependency and the first-run model download are handled correctly.

2. Testing

2.1 Testing Approach

Level	What is tested	Why it is useful
Unit	Each module alone (explanation_module, qna, summary_module, quiz_module, learning_path), including empty input	Finds mistakes in one feature without involving the web layer
Integration	POST /generate routes each task to the right module and shows the right model label	Confirms the pieces are connected correctly
Functional (UI)	The page in a browser: drop-down, textarea, submit button, answer box, badge	Confirms the student experience
Error and robustness	Missing or wrong key, API failure, invalid task, empty input	Confirms the page still renders with a clear message

2.2 Test Environment

Item	Details
Framework	Python 3, FastAPI, Jinja2
Server	Uvicorn
Local model	MBZUAI/LaMini-Flan-T5-783M (Transformers + PyTorch, CPU)
Cloud model	Gemini Flash through google-genai; key in GEMINI_API_KEY
Browser	Any modern browser such as Chrome, Edge or Firefox

2.3 Test Cases

ID	Test case	Input	Expected result
TC-01	Home page	GET /	index.html loads; Concept selected; no answer box
TC-02	Concept	task=concept, "Photosynthesis"	Short explanation; badge shows LaMini-Flan-T5-783M (Local CPU)
TC-03	Q&A	task=qa, "What is recursion?"	Detailed answer with examples; Gemini badge
TC-04	Summary	task=summary, a paragraph	1-2 sentence overview and bullet takeaways

ID	Test case	Input	Expected result
TC-05	Quiz	task=quiz, "Python lists"	3 questions with options A-D and an Answer Key
TC-06	Learning path	task=path, "Machine Learning"	Phased roadmap with topics, projects and timelines
TC-07	Empty concept input	task=concept, spaces only	"Please provide a concept or topic to explain."
TC-08	Empty Gemini input	qa / summary / quiz / path, spaces only	Task-specific "Please provide ..." message; no API call
TC-09	Missing field	POST without prompt	422 validation error
TC-10	Invalid task	task="unknown"	422 error from the TaskType enum
TC-11	No API key	Start without GEMINI_API_KEY	RuntimeError at start-up
TC-12	OAuth token as key	Key starting with ya29.	RuntimeError asking for an AI Studio key
TC-13	API failure	Invalid key or no network	"Error generating response: ..." and badge System Error
TC-14	First model load	First concept request	Model loads once; later requests are faster
TC-15	Long input	Concept prompt over 512 tokens	Input truncated to 512 tokens; output at most 256 new tokens
TC-16	Form state kept	Submit any request	Selected task and query are still shown after the page reloads
TC-17	Output safety	Prompt that makes the model output HTML tags	Tags are shown as text because Jinja2 escapes the response

2.4 Automated Route Tests

Route tests can be automated with pytest and FastAPI's TestClient. The Gemini function is replaced with a mock, so tests run without using the API quota.

```
pip install pytest httpx

# test_app.py
from fastapi.testclient import TestClient
from unittest.mock import patch
import Main

client = TestClient(Main.app)

def test_home():
    assert client.get('/').status_code == 200

def test_invalid_task():
    r = client.post('/generate', data={'prompt': 'x', 'task': 'bad'})
    assert r.status_code == 422

def test_qa_routing():
    with patch('Main.qna.get_answer', return_value='mock answer'):
        r = client.post('/generate', data={'prompt': 'hi', 'task': 'qa'})
    assert 'mock answer' in r.text

# run:  pytest -v
```

Importing Main requires a GEMINI_API_KEY value (it can be a dummy string for route tests) and existing templates and static folders, because Main.py checks the key and mounts those folders when it is imported.

2.5 Observations and Limitations

- The local model gives short answers, so its quality is lower than the Gemini tasks.
- Gemini output varies between runs, so tests should check structure (for example the presence of an Answer Key) rather than exact wording.
- The first concept request is slow because the model loads into memory.
- All four cloud tasks depend on network access and API quota.

3. Deployment

3.1 Requirements

Requirement	Description
Python	3.9 or higher
Packages	Everything in requirements.txt
API key	Gemini key from Google AI Studio in the .env file or an environment variable
Memory	Around 4 GB free is recommended for PyTorch and the local model (estimate)
Network	Internet for Gemini calls and the one-time Hugging Face model download

3.2 Project Structure

```
EduGenie/
|-- Main.py
|-- explanation_module.py
|-- qna.py
|-- summary_module.py
|-- quiz_module.py
|-- learning_path.py
|-- gemini_helper.py
|-- templates/index.html
|-- static/style.css
|-- requirements.txt
|-- .env                (never commit this file)
```

3.3 Local Deployment

- Create and activate a virtual environment.
- Run pip install -r requirements.txt.
- Create .env containing GEMINI_API_KEY=your_api_key_here.
- Start the server: uvicorn Main:app --host 127.0.0.1 --port 8000 --reload
- Open http://127.0.0.1:8000 in the browser.

The file is named Main.py with a capital M, so the Uvicorn target must be written Main:app. On case-sensitive systems such as Linux, main:app would not be found.

3.4 Production Deployment

- Run without --reload, for example: uvicorn Main:app --host 0.0.0.0 --port 8000 --workers 2. Each worker loads its own copy of the local model, so memory grows with the worker count.
- Put Nginx or another reverse proxy in front for HTTPS and static files.
- Provide the API key as an environment variable or platform secret instead of a file in the repository.
- Pre-download the model during the build step, or keep a persistent cache, so the first user does not wait.
- Run the process under a supervisor such as systemd so it restarts after a crash.

```
FROM python:3.11-slim
WORKDIR /app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY . .
EXPOSE 8000
CMD ["uvicorn", "Main:app", "--host", "0.0.0.0", "--port", "8000"]

# docker build -t edugenie .
# docker run -p 8000:8000 -e GEMINI_API_KEY=your_key edugenie
```

3.5 Post-Deployment Checklist

- The home page opens and style.css loads (the card is centred and styled).
- One request for each of the five tasks works and shows the correct badge.
- Logs show the local model loading once without errors.
- The API key does not appear in the repository, logs or page source.
- HTTPS is active and the server restarts automatically after a failure.